

		Location		
Factor	Weight	A	B	C
Easy access	0.15	86	72	90
Parking facilities	0.20	72	77	91
Display area	0.15	86	90	90
Shopper (walking) traffic	0.10	94	86	80
Transportation costs	0.15	99	89	81
Operating costs	0.25	96	85	75

Industrial Facilities Design
Course Code: EF431

		Location		
		Score		
Factor	Weight	A	B	C
Easy Access	0.15	86	72	90
Parking Facilities	0.2	72	77	91
Display area	0.15	86	90	90
Shopper (walking) traffic	0.1	94	86	80
Transportation costs	0.15	99	89	81
Operating costs	0.25	96	85	75
Total Weight Sum ->>>>		1		

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a. Using the factor ratings shown below, determine which location alternative should be chosen on the basis of maximum composite score.

		Location			Composite Score		
		Score					
Factor	Weight	A	B	C	A	B	C
	(1)	(2)	(3)	(4)	(1)x(2)	(1)x(3)	(1)x(4)
Easy Access	0.15	86	72	90	12.9	10.8	13.5
Parking Facilities	0.2	72	77	91	14.4	15.4	18.2
Display area	0.15	86	90	90	12.9	13.5	13.5
Shopper (walking) traffic	0.1	94	86	80	9.4	8.6	8
Transportation costs	0.15	99	89	81	14.85	13.35	12.15
Operating costs	0.25	96	85	75	24	21.25	18.75
Total Weight Sum ->>>>	1	Total Score ->>>>			88.45	82.9	84.1

Decision Final

88.45

b. If transportation costs and operating costs are given weights that are double the weights of the others, should the locations be ranked differently?

Solution:

Let the weight of the other factors be w .

Weights:

1. Easy access: w
2. Parking facilities: w
3. Display area: w
4. Shopper traffic: w
5. Transportation costs: $2w$
6. Operating costs: $2w$

The sum of all weights must be 1.

$$w + w + w + w + 2w + 2w = 1$$

$$4w + 4w = 1$$

$$8w = 1$$

$$w = \frac{1}{8} = 0.125$$

The new weights are:

Four factors @ 0.125
Two factors @ $2w = 0.25$

		Location					
		Score			Composite Score		
Factor	Weight	A	B	C	A	B	C
	(1)	(2)	(3)	(4)	(1)x(2)	(1)x(3)	(1)x(4)
Easy Access	0.125	86	72	90	10.75	9	11.25
Parking Facilities	0.125	72	77	91	9	9.625	11.375
Display area	0.125	86	90	90	10.75	11.25	11.25
Shopper (walking) traffic	0.125	94	86	80	11.75	10.75	10
Transportation costs	0.25	99	89	81	24.75	22.25	20.25
Operating costs	0.25	96	85	75	24	21.25	18.75
Total Weight Sum ->>>>	1	Total Score ->>>>			91	84.125	82.875

Decision Final

91

c. If the manager weights the factors equally, how would the locations be ranked?

If all 6 factors are weighted equally, each factor has a weight of $\frac{1}{6}$. To find the ranking, we can simply sum the raw scores for each location. The location with the highest total score will be ranked first.

		Location					
		Score			Composite Score		
Factor	Weight	A	B	C	A	B	C
	(1)	(2)	(3)	(4)	(1)x(2)	(1)x(3)	(1)x(4)
Easy Access	0.166667	86	72	90	14.33333	12	15
Parking Facilities	0.166667	72	77	91	12	12.83333	15.16667
Display area	0.166667	86	90	90	14.33333	15	15
Shopper (walking) traffic	0.166667	94	86	80	15.66667	14.33333	13.33333
Transportation costs	0.166667	99	89	81	16.5	14.83333	13.5
Operating costs	0.166667	96	85	75	16	14.16667	12.5
Total Weight Sum ->>>>	1	Total Score ->>>>			88.83333	83.16667	84.5

Decision Final

88.83333